

		0	00	
1 to 18 EVEN	1st Dozen	1	2	3
		4	5	6
		7	8	9
	2nd Dozen	10	11	12
		13	14	15
		16	17	18
	3rd Dozen	19	20	21
		22	23	24
		25	26	27
ODD 19 to 36		28	29	30
		31	32	33
		34	35	36
		2 to 1	2 to 1	2 to 1

Roulette Table (credit: "American Roulette Table Layout" by Film8ker/Wikimedia Commons, Public Domain)

29. If a player bets \$1 on red and wins, the player gets \$2 back (the original \$1 bet plus \$1 winnings). What is the probability that the player wins more than they lose if the player bets on red on 5 consecutive spins?
30. What is the probability that the player wins more than they lose if the player bets on red on 15 consecutive spins?
31. What is the probability that the player wins more than they lose if the player bets on red on 30 consecutive spins?
32. What is the probability that the player wins more than they lose if the player bets on red on 100 consecutive spins?
33. What is the probability that the player wins more than they lose if the player bets on red on 200 consecutive spins?
34. What is the probability that the player wins more than they lose if the player bets on red on 1,000 consecutive spins?
35. What is the probability that the player wins more than they lose if the player bets on red on 5,000 consecutive spins?

7.11 Expected Value



Figure 7.46 The concept of expected value allows us to analyze games that involve randomness, like Roulette. (credit: "Roulette Table and Roulette Wheel in a Casino with People betting on numbers" by Marco Verch/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Calculate the expected value of an experiment.
2. Interpret the expected value of an experiment.
3. Use expected value to analyze applications.

The casino game roulette has dozens of different bets that can be made. These bets have different probabilities of

winning but also have different payouts. In general, the lower the probability of winning a bet is, the more money a player wins for that bet. With so many options, is there one bet that's "smarter" than the rest? What's the best play to make at a roulette table? In this section, we'll develop the tools we need to answer these questions.

Expected Value

Many experiments have numbers associated with their outcomes. Some are easy to define; if you roll 2 dice, the sum of the numbers showing is a good example. In some card games, cards have different point values associated with them; for example, in some forms of the game rummy, aces are worth 15 points; 10s, jacks, queens, and kings are worth 10; and all other cards are worth 5. The outcomes of casino and lottery games are all associated with an amount of money won or lost. These outcome values are used to find the **expected value** of an experiment: the mean of the values associated with the outcomes that we would observe over a large number of repetitions of the experiment. (See [Conditional Probability and the Multiplication Rule](#) for more on means.)

That definition is a little vague; How many is "a large number?" In practice, it depends on the experiment; the number has to be large enough that every outcome would be expected to appear at least a few times. For example, if we're talking about rolling a standard 6-sided die and we note the number showing, a few dozen replications should be enough that the mean would be representative. Since the probability of each outcome is $\frac{1}{6}$, we would expect to see each outcome about 8 times over the course of 48 replications. However, if we're talking about the Powerball lottery, where the probability of winning the jackpot is about $\frac{1}{292,000,000}$, we would need several *billion* replications to ensure that every outcome appears a few times. Luckily, we can find the theoretical expected value before we even run the experiment the first time.

FORMULA

Expected Value: If O represents an outcome of an experiment and $n(O)$ represents the value of that outcome, then the expected value of the experiment is:

$$\sum n(O) \times P(O),$$

where Σ is the "sum," meaning we add up the results of the formula that follows over all possible outcomes.

EXAMPLE 7.37

Finding Expected Values

Find the expected values of the following experiments.

1. Roll a standard 6-sided die and note the number showing.
2. Roll two standard 6-sided dice and note the sum of the numbers showing.
3. Draw a card from a well-shuffled standard deck of cards and note its rummy value (15 for aces; 10 for tens, jacks, queens, and kings; 5 for everything else).

Solution

1. **Step 1:** Let's start by writing out the PDF table for this experiment.

Value	Probability
1	$\frac{1}{6}$
2	$\frac{1}{6}$
3	$\frac{1}{6}$
4	$\frac{1}{6}$

Value	Probability
5	$\frac{1}{6}$
6	$\frac{1}{6}$

Step 2: To find the expected value, we need to find $n(O) \times P(O)$ for each possible outcome in the table below.

Value	Probability	$n(O) \times P(O)$
1	$\frac{1}{6}$	$1 \times \frac{1}{6} = \frac{1}{6}$
2	$\frac{1}{6}$	$2 \times \frac{1}{6} = \frac{1}{3}$
3	$\frac{1}{6}$	$3 \times \frac{1}{6} = \frac{1}{2}$
4	$\frac{1}{6}$	$4 \times \frac{1}{6} = \frac{2}{3}$
5	$\frac{1}{6}$	$5 \times \frac{1}{6} = \frac{5}{6}$
6	$\frac{1}{6}$	$6 \times \frac{1}{6} = 1$

Step 3: We add all of the values in that last column: $\frac{1}{6} + \frac{1}{3} + \frac{1}{2} + \frac{2}{3} + \frac{5}{6} + 1 = \frac{7}{2} = 3.5$. So, the expected value of a single roll of a die is 3.5.

2. Back in [Example 7.18](#), we made this table of all of the equally likely outcomes ([Figure 7.47](#)):

		First Die					
		1	2	3	4	5	6
Second Die	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

Figure 7.47

Step 1: Let's use [Figure 7.47](#) to create the PDF for this experiment, as shown in the following table:

Value	Probability
2	$\frac{1}{36}$
3	$\frac{1}{18}$
4	$\frac{1}{12}$
5	$\frac{1}{9}$
6	$\frac{5}{36}$
7	$\frac{1}{6}$
8	$\frac{5}{36}$
9	$\frac{1}{9}$
10	$\frac{1}{12}$
11	$\frac{1}{18}$
12	$\frac{1}{36}$

Step 2: We can multiply each row to find $n(O) \times P(O)$ as shown in the following table:

Value	Probability	$n(O) \times P(O)$
2	$\frac{1}{36}$	$\frac{1}{18}$
3	$\frac{1}{18}$	$\frac{1}{6}$
4	$\frac{1}{12}$	$\frac{1}{3}$
5	$\frac{1}{9}$	$\frac{5}{9}$
6	$\frac{5}{36}$	$\frac{5}{6}$
7	$\frac{1}{6}$	$\frac{7}{6}$
8	$\frac{5}{36}$	$\frac{10}{9}$
9	$\frac{1}{9}$	1
10	$\frac{1}{12}$	$\frac{5}{6}$

Value	Probability	$n(O) \times P(O)$
11	$\frac{1}{18}$	$\frac{11}{18}$
12	$\frac{1}{36}$	$\frac{1}{3}$

Step 3: We can add the last column to get the expected value:

$$\frac{1}{18} + \frac{1}{6} + \frac{1}{3} + \frac{5}{9} + \frac{5}{6} + \frac{7}{6} + \frac{10}{9} + 1 + \frac{5}{6} + \frac{11}{18} + \frac{1}{3} = 7.$$

So, the expected value is 7.

3. **Step 1:** Let's make a PDF table for this experiment. There are 3 events that we care about, so let's use those events in the table below:

Event	Probability
{A}	$\frac{1}{13}$
{10, J, Q, K}	$\frac{4}{13}$
{2, 3, 4, 5, 6, 7, 8, 9}	$\frac{8}{13}$

Step 2: Let's add a column to the following table for the values of each event:

Event	Probability	Value
{A}	$\frac{1}{13}$	15
{10, J, Q, K}	$\frac{4}{13}$	10
{2, 3, 4, 5, 6, 7, 8, 9}	$\frac{8}{13}$	5

Step 3: We'll add the column for the product of the values and probabilities to the table below:

Event	Probability	Value	$n(O) \times P(O)$
{A}	$\frac{1}{13}$	15	$\frac{15}{13}$
{10, J, Q, K}	$\frac{4}{13}$	10	$\frac{40}{13}$
{2, 3, 4, 5, 6, 7, 8, 9}	$\frac{8}{13}$	5	$\frac{40}{13}$

Step 4: We'll find the sum of the last column: $\frac{15}{13} + \frac{40}{13} + \frac{40}{13} = \frac{95}{13} \approx 7.3$. Thus, the expected *Rummy* value of a randomly selected card is about 7.3.

> YOUR TURN 7.37

1. Find the expected value of the number showing when you roll a special 6-sided die with faces {1, 1, 2, 3, 5, 8}.
2. Find the expected value of the number of heads showing if you flip a coin 3 times.
3. You are about to play a game, where you flip a coin 3 times. If all 3 flips result in heads, you win \$20. If you get 2 heads, you win \$10. If you flip 1 or 0 heads, you win nothing. What is the expected value of your winnings?

Let's make note of some things we can learn from [Example 7.37](#). First, as Exercises 1 and 3 demonstrate, the expected value of an experiment might not be a value that could come up in the experiment. Remember that the expected value is interpreted as a mean, and the mean of a collection of numbers doesn't have to actually be one of those numbers.

Second, looking at Exercise 1, the expected value (3.5) was just the mean of the numbers on the faces of the die:

$\frac{1+2+3+4+5+6}{6} = 3.5$. This is no accident! If we break that fraction up using the addition in the numerator, we get $\frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6}$, which we can rewrite as $1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$. That's exactly the computation we did to find the expected value! In fact, expected values can always be treated as a special kind of mean called a **weighted mean**, where the weights are the probabilities associated with each value. When the probabilities are all equal, the weighted mean is just the regular mean.

Interpreting Expected Values

As we noted, the expected value of an experiment is the mean of the values we would observe if we repeated the experiment a large number of times. (This interpretation is due to an important theorem in the theory of probability called the Law of Large Numbers.) Let's use that to interpret the results of the [previous example](#).

EXAMPLE 7.38

Interpreting Expected Values

Interpret the expected values of the following experiments.

1. Roll a standard 6-sided die and note the number showing.
2. Roll 2 standard 6-sided dice and note the sum of the numbers showing.
3. Draw a card from a well-shuffled standard deck of cards and note its *Rummy* value (15 for aces; 10 for tens, jacks, queens, and kings; 5 for everything else).

✓ Solution

1. If you roll a standard 6-sided die many times, the mean of the numbers you roll will be around 3.5.
2. If you roll a pair of standard 6-sided dice many times, the mean of the sums of the numbers you roll will be about 7.
3. If you draw a card from a well-shuffled deck many times, the mean of the *Rummy* values of the cards would be around 7.3.

> YOUR TURN 7.38

1. Interpret the expected value of the number showing when you roll a special 6-sided die with faces {1, 1, 2, 3, 5, 8}.
2. Interpret the expected value of the number of heads showing if you flip a coin 3 times.
3. You are about to play a game where you flip a coin 3 times. If all 3 flips result in heads, you win \$20. If you get 2 heads, you win \$10. If you flip 1 or 0 heads, you win nothing. Interpret the expected value of your winnings.

?? WHO KNEW?

Pascal's Wager

The French scholar Blaise Pascal (1623–1662) was among the earliest mathematicians to study probabilities, and was the first to accurately describe and compute expected values. In his book *Pensées (Thoughts)*, he turned the analysis of expected values to his belief in the Christian God. He said that there is no way for people to establish the

probability that God exists, but since the “winnings” on a bet that God exists (and that you then lead your life accordingly) are essentially infinite, the expected value of taking that bet is always positive, no matter how unlikely it is that God exists.

Using Expected Value

Now that we know how to find and interpret expected values, we can turn our attention to using them. Suppose someone offers to play a game with you. If you roll a die and get a 6, you get \$10. However, if you get a 5 or below, you lose \$1. Is this a game you'd want to play? Let's look at the expected value: The probability of winning is $\frac{1}{6}$ and the probability of losing is $\frac{5}{6}$, so the expected value is $\$10 \times \frac{1}{6} + (-\$1) \times \frac{5}{6} = \frac{\$5}{6} \approx \$0.83$. That means, on average, you'll come out ahead by about 83 cents every time you play this game. It's a great deal! On the other hand, if the winnings for rolling a 6 drop to \$3, the expected value becomes $\$3 \times \frac{1}{6} + (-\$1) \times \frac{5}{6} = -\frac{\$1}{3} \approx -\$0.33$, meaning you should expect to lose about 33 cents on average for every time you play. Playing that game is not a good idea! In general, this is how casinos and lottery corporations make money: Every game has a negative expected value for the player.

? WHO KNEW?

Expected Values in Football

In the 21st century, data analytics tools have revolutionized the way sports are coached and played. One tool in particular is used in football at crucial moments in the game. When a team faces a fourth down (the last possession in a series of four possessions, a fairly common occurrence), the coach faces a decision: Run one play to try to gain a certain number of yards, or kick the ball away to the other team. Here's the interesting part of the decision: If the team “goes for it” and runs the play and they are successful, then they keep possession of the ball and can continue in their quest to score more points. If they are unsuccessful, then they lose possession of the ball, giving the other team an opportunity to score points. If, instead, the team punts, or kicks the ball away, then the other team gets possession of the ball, but in a worse position for them than if the original team goes for it and fails. To analyze this situation, data analysts have generated empirical probabilities for every fourth down situation, and computed the expected value (in terms of points) for each decision. Coaches frequently use those calculations when they decide which option to take!



PEOPLE IN MATHEMATICS

Pierre de Fermat and Blaise Pascal

In 1654, a French writer and amateur mathematician named Antoine Gombaud (who called himself the Chevalier du Mére) reached out to his gambling buddy Blaise Pascal to answer a question that he'd read about called the “problem of points.” The question goes like this: Suppose you're playing a game that is scored using points, and the first person to earn 5 points is the winner. The game is interrupted with the score 4 points to 2. If the winner stood to win \$100, how should the prize money be divided between the players? Certainly the person who is 1 point away from victory should get more, but *how much* more?

We have developed tools in this section to answer this question. At its heart, it's a question about conditional probabilities and expected value. At the time that Pascal first started thinking about it, though, those ideas hadn't yet been invented. Pascal reached out to a colleague named Pierre de Fermat, and over the course of a couple of months, their correspondence with each other would eventually solve the problem. In the process, they first described conditional probabilities and expected values!

Apart from their work in probability, these men are famous for other work in mathematics (and, in Pascal's case, philosophy and physics). Fermat is remembered for his work in geometry and in number theory. After his death, the statement of what came to be called “Fermat's Last Theorem” was discovered scribbled in the margin of a book, with the note that Fermat had discovered a “marvelous proof that this margin is too small to contain.” The theorem says that any equation of the form $a^n + b^n = c^n$ has no positive integer solutions if $n \geq 3$. No proof of that theorem was

discovered until 1994, when Andrew Wiles used computers and new branches of geometry to finally prove the theorem!

Pascal is remembered for the “arithmetical triangle” that is named for him (though he wasn't the first person to discover it; see the [section on the binomial distribution](#) for more), as well as work in geometry. In physics, Pascal worked on hydrodynamics and air pressure (the SI unit for pressure is named for him), and in philosophy, Pascal advocated for a mathematical approach to philosophical problems.

EXAMPLE 7.39

Using Expected Values

In the casino game keno, a machine chooses at random 20 numbers between 1 and 80 (inclusive) without replacement. Players try to predict which numbers will be chosen. Players don't try to guess all 20, though; generally, they'll try to predict between 1 and 10 of the chosen numbers. The amount won depends on the number of guesses they made and the number of guesses that were correct.

1. At one casino, a player can try to guess just 1 number. If that number is among the 20 selected, the player wins \$2; otherwise, the player loses \$1. What is the expected value?
2. At the same casino, if a player makes 2 guesses and they're both correct, the player wins \$14; otherwise, the player loses \$1. What is the expected value?
3. Players can also make 3 guesses. If 2 of the 3 guesses are correct, the player wins \$1. If all 3 guesses are correct, the player wins \$42. Otherwise, the player loses \$1. What is the expected value?
4. Which of these games is the best for the player? Which is the best for the casino?

Solution

1. There are 20 winning numbers out of 80, so if we try to guess one of them, the probability of guessing correctly is $\frac{20}{80} = \frac{1}{4}$. The probability of losing is then $\frac{3}{4}$, and so the expected value is $\$2 \times \frac{1}{4} + (-\$1) \times \frac{3}{4} = -\0.25 .
2. There are ${}_{20}C_2 = 190$ winning choices out of ${}_{80}C_2 = 3,160$ total ways to choose 2 numbers. So, the probability of winning is $\frac{190}{3,160}$ and the probability of losing is $\frac{3,160-190}{3,160} = \frac{2,970}{3,160}$. So, the expected value of the game is $\$14 \times \frac{190}{3,160} + (-\$1) \times \frac{2,970}{3,160} \approx -\0.10 .
3. **Step 1:** Let's start with the big prize. There are ${}_{20}C_3 = 1,140$ ways to correctly guess 3 winning numbers out of ${}_{80}C_3 = 82,160$ ways to guess three numbers total. That means the probability of winning the big prize is $\frac{1,140}{82,160} \approx 0.01388$.

Step 2: Let's find the probability of the second prize. The denominator is the same: 82,160. Let's figure out the numerator. To win the second prize, the player must pick 2 of the 20 winning numbers and one of the 60 losing numbers. The number of ways to do that can be found using the Multiplication Rule for Counting: there are ${}_{20}C_2 = 190$ ways to pick 2 winning numbers and 60 ways to pick 1 losing number, so there are $190 \times 60 = 11,400$ ways to win the second prize. So, the probability of winning that second prize is $\frac{11,400}{82,160} \approx 0.13875$.

Step 3: Since the overall probability of winning is $\frac{1,140}{82,160} + \frac{11,400}{82,160} = \frac{12,540}{82,160} \approx 0.15263$, the probability of losing must be $1 - 0.15263 = 0.84737$. So, the expected value is $\$42 \times 0.01388 + \$1 \times 0.13875 + (-\$1) \times 0.84737 \approx -\0.13 .

4. The bet that's the best for the player is the one with the highest expected value for the player, which is guessing two numbers. The best one for the casino is the one with the lowest expected value for the player, which is guessing one number.

YOUR TURN 7.39

The casino game craps involves rolling 2 standard 6-sided dice. While the main game involves repeated rolls of the dice, players can also bet on the outcomes of single rolls. Find the expected values of the following three \$1 bets. Then decide which bet is the best for the player and which bet is the best for the casino.

1. Players can bet on the next roll of the dice being a sum of 7. Winners get \$4 and losers lose \$1.
2. Players can bet on the next roll of the dice being a sum of 12. Winners get \$30 and losers lose \$1

3. Players can bet that the next roll of the dice will be “any craps” (a sum of 2, 3, or 12). Winners get \$7 and losers lose \$1.

WORK IT OUT

Make Your Own Lottery

By yourself or with a partner, devise your own lottery scheme. Assume you would have access to one or more machines that choose numbers randomly. What will a lottery draw look like? How many numbers are players choosing from? How many will be drawn? Will they be drawn with replacement or without replacement? What conditions must be met for a player to win first or second (or more!) prize? Once you’ve decided that, decide the payoff structure for winners, and how much the game will cost to play. Try to make the game enticing enough that people will want to play it, but with enough negative expected value that the lottery will make money. Aim for the expected value to be about -0.25 times the cost of playing the game.

Check Your Understanding

You are about to roll a 20-sided die with faces labeled as follows: 5 faces have a 1, 6 faces have a 3, 4 faces have a 5, 3 faces have a 7, and 2 faces have a 9.

62. What is the expected value of the number showing on the die after it’s rolled?
63. Interpret your answer.

You are about to play a game in which you draw 3 ping-pong balls without replacement from a barrel. The barrel contains 6 green balls and 4 red balls. If all 3 of your selections are green, you win \$5. If 2 of the 3 are green, you win \$1. If 2 or more of your selections are red, you lose \$5.

64. What is the expected value of this game?
65. Interpret your answer.
66. Is it advantageous to you to play the game? How do you know?



SECTION 7.11 EXERCISES

You roll a standard 6-sided die and win points equal to the square of the number shown.

1. What’s the expected value of the number of points you win?
2. Interpret your answer.

In the classic board game *The Game of Life*, players have the chance to play the market. A spinner with 10 equally likely spaces is spun to choose a random number. If the result is 3 or less, the player loses \$25,000. If the result is 7 or more, the player wins \$50,000. If the result is a 4, 5, or 6, the player doesn’t win or lose anything.

3. What is the expected value of playing the market?
4. Interpret the answer.

The Game of Life players also occasionally have the opportunity to speculate. Players choose any 2 of the 10 numbers on the spinner and then give it a spin. If one of their numbers is chosen, they win \$140,000; if not, they lose \$10,000.

5. What is the expected value of this speculation?
6. Interpret your answer.

7. Which is better for *The Game of Life* players: playing the market or speculating? How do you know?

A charitable organization is selling raffle tickets as a fundraiser. They intend to sell 5,000 tickets at \$10 each. One ticket will be randomly selected to win the grand prize of a new car worth \$35,000.

8. What is the expected value of a single ticket?
9. Interpret your answer.
10. The organization is worried they won’t be able to sell all the tickets, so they announce that, in addition to the grand prize, they will offer 10 second prizes of \$500 in cash. What is the new expected value of a single ticket?
11. Interpret your answer.

In the following exercises involve randomly selecting golf balls from a bucket. The bucket contains 4 yellow balls (numbered 1-4) and 6 white balls (numbered 1-6).

12. If you draw a single ball, what is the expected number of yellow balls selected?

13. Suppose you draw 2 balls *with* replacement.
 - a. Give a PDF table for the possible outcomes for the number of yellow balls selected.
 - b. What is the expected number of yellow balls selected?
14. Suppose you draw 2 balls *without* replacement.
 - a. Give a PDF table for the possible outcomes for the number of yellow balls selected.
 - b. What is the expected number of yellow balls selected?
15. Suppose you draw 3 balls *with* replacement.
 - a. Give a PDF table for the possible outcomes for the number of yellow balls selected.
 - b. What is the expected number of yellow balls selected?
16. Suppose you draw 3 balls *without* replacement.
 - a. Give a PDF table for the possible outcomes for the number of yellow balls selected.
 - b. What is the expected number of yellow balls selected?
17. If you draw a single ball, what is the expected value of the number on the ball?
18. Suppose you draw 2 balls *with* replacement.
 - a. Give a PDF table for the possible outcomes for the sum of the numbers on the selected balls.
 - b. What is the expected sum of the numbers on the balls?
19. Suppose you draw 2 balls *without* replacement.
 - a. Give a PDF table for the possible outcomes for the sum of the numbers on the selected balls.
 - b. What is the expected sum of the numbers on the balls?

The following exercises deal with the game “Punch a Bunch,” which appears on the TV game show *The Price Is Right*. In this game, contestants have a chance to punch through up to 4 paper circles on a board; behind each circle is a card with a dollar amount printed on it. There are 50 of these circles; the dollar amounts are given in this table:

Dollar Amount	Frequency
\$25,000	1
\$10,000	2
\$5,000	4
\$2,500	8
\$1,000	10
\$500	10
\$250	10
\$100	5

Contestants are shown their selected dollar amounts one at a time, in the order selected. After each is revealed, the contestant is given the option of taking that amount of money or throwing it away in favor of the next amount. (You can watch the game being played in the video [Playing "Punch a Bunch."](https://openstax.org/r/Playing_Punch_a_Bunch) (https://openstax.org/r/Playing_Punch_a_Bunch)) Anita is playing “Punch a Bunch” and gets 2 punches.

20. If Anita got \$500 on her first punch, what's the expected value of her second punch?
21. If Anita got \$500 on her first punch, should she throw out her \$500 and take the results of her second punch? How do you know?
22. If Anita got \$1,000 on her first punch, what's the expected value of her second punch?
23. If Anita got \$1,000 on her first punch, should she throw out her \$1,000 and take the results of her second punch? How do you know?
24. If Anita got \$2,500 on her first punch, what's the expected value of her second punch?

25. If Anita got \$2,500 on her first punch, should she throw out her \$2,500 and take the results of her second punch? How do you know?

The following exercises are about the casino game roulette. In this game, the dealer spins a marble around a wheel that contains 38 pockets that the marble can fall into. Each pocket has a number (each whole number from 0 to 36, along with a “double zero”) and a color (0 and 00 are both green; the other 36 numbers are evenly divided between black and red). Players make bets on which number (or groups of numbers) they think the marble will land on. The figure shows the layout of the numbers and colors, as well as some of the bets that can be made.

		0	00
1 to 18 EVEN	1st Dozen	1	2
		4	5
		7	8
		10	11
2nd Dozen		13	14
		16	17
		19	20
		22	23
3rd Dozen		25	26
		28	29
		31	32
		34	35
		36	
		2 to 1	2 to 1

Roulette Table (credit: "American Roulette Table Layout" by Film8ker/Wikimedia Commons, Public Domain)

26. If a player makes a \$1 bet on a single number, they win \$35 if that number comes up, but lose \$1 if it doesn't. What is the expected value of this bet?
27. Interpret your answer to the previous question.
28. Suppose a player makes the \$1 bet on a single number in 5 consecutive spins. What is the expected value of this series of bets? (Hint: use the Binomial Distribution.)
29. Interpret your answer to the previous question.
30. If a player makes a \$10 bet on first dozen, they win \$20 if one of the numbers 1–12 comes up but lose \$10 otherwise. What is the expected value of this bet?
31. Interpret your answer to the previous question.
32. Suppose a player makes the \$10 bet on first dozen in 4 consecutive spins. What is the expected value of that series of bets?
33. Interpret your answer to the previous question.
34. If a player makes a \$10 basket bet, they win \$60 if 0, 00, 1, 2, or 3 come up but lose \$10 otherwise. What is the expected value of this bet?
35. Interpret your answer to previous question.
36. Which is better for the player: a \$10 first dozen bet or a \$10 basket bet? How do you know?